EE 471C / EE 381K-17 Wireless Communication Lab

Preview of Today's Lecture

- Explain different impairments caused by the frequency flat wireless communication channel
  - Delay
  - Symbol timing mismatch
  - Frame timing mismatchs
- Formulate the discrete-time symbol synchronization problem
- Solve the symbol synchronization problem using maximum output energy maximization

Impairments in frequency flat channels

Learning objectives:

- Explain different impairments caused by the frequency flat wireless communication channel

Frequency flat wireless channel model

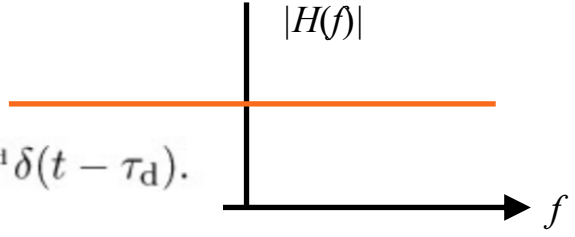
$$h(\tau) = \alpha e^{j\phi}\delta(t-\tau_d)$$

Single-path channel

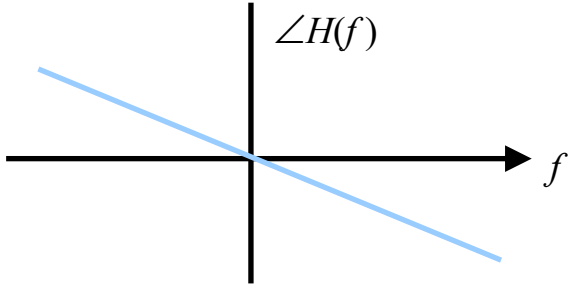
- Baseband equivalent channel

$$h(t) = B\alpha e^{-\mathrm{j}2\pi f_c \tau_d}\mathrm{sinc}(B(t-\tau_d))$$

  - Pseudo-baseband equivalent $h_{\mathrm{pb}}(t) = \alpha e^{-\mathrm{j}2\pi f_c \tau_d}\delta(t-\tau_d).$



  - Frequency domain $H(f) = \alpha e^{-j2\pi\tau_d + j\phi}$
  - Frequency flat because |H(f)| is constant



Channel captures the impact of propagation delay

Sampled received signal

- At the receiver prior to matched filtering

$$r(t) = \sqrt{E_{\mathrm{x}}} h(t) * g_{\mathrm{tx}}(t) * \sum_m s[m]\delta(t - mT) + v(t)$$

$$= \sqrt{E_{\mathrm{x}}} \alpha e^{-\mathrm{j}2\pi f_{\mathrm{c}} \tau_{\mathrm{d}}} \sum_m s[m] g_{\mathrm{tx}}(t - mT - \tau_{\mathrm{d}}) + v(t)$$

$$= h \sum_m s[m] g_{\mathrm{tx}}(t - mT - \tau_{\mathrm{d}}) + v(t)$$

- After sampling

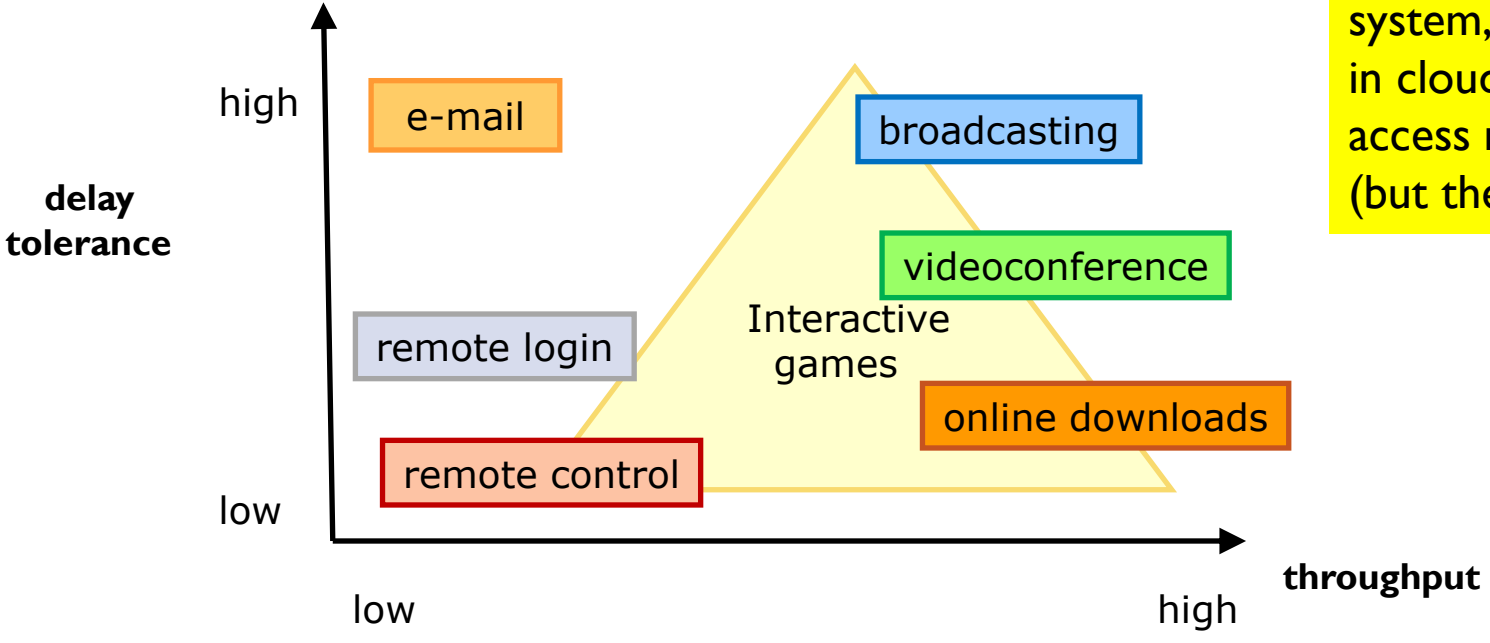
Unknown channel

Unknown delay

$$y[n] = h \sum_{m=-\infty}^{\infty} s[m] g((n-m)T - \tau_{\mathrm{d}}) + v[n]$$

Latency and delay in wireless systems

- Different applications support different latencies



Propagation delay is rarely the dominant source of latency in a system, sometimes in cloud radio access networks (but the network)

Motivation for symbol synchronization

- Consider the sampled signal after matched filtering

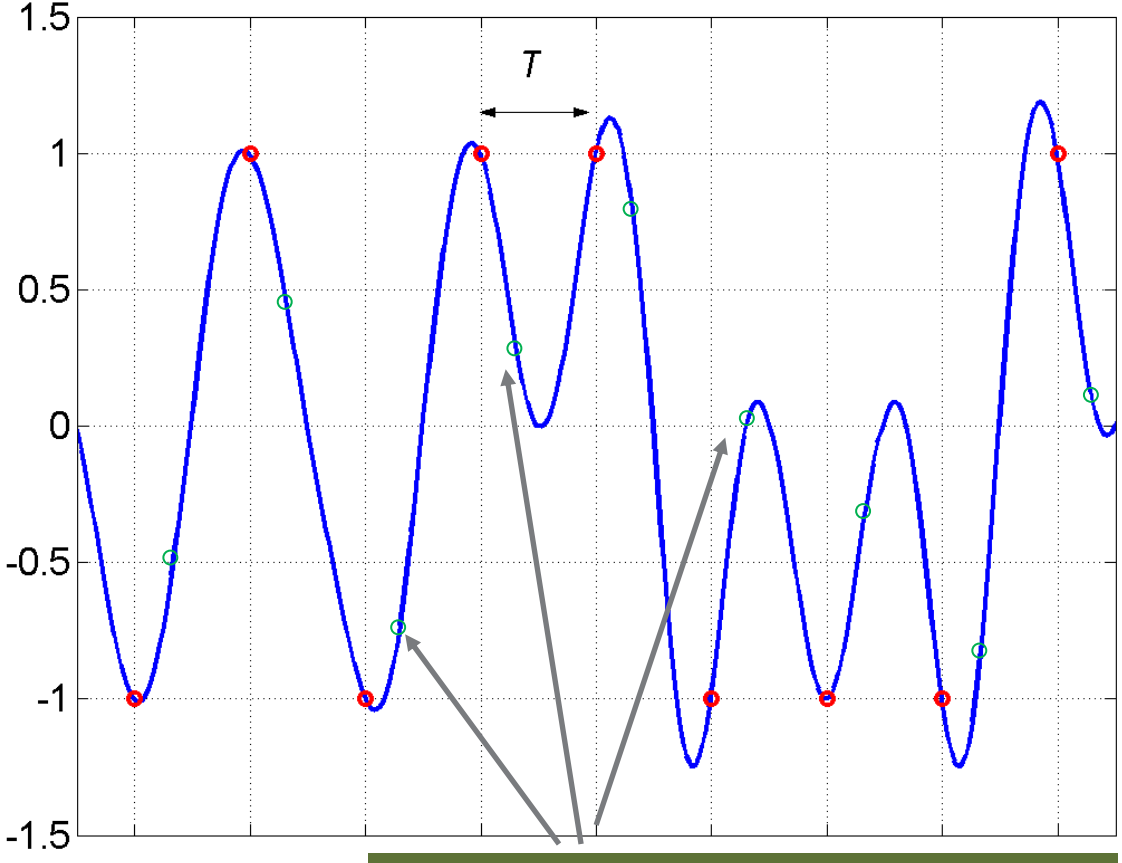
$$y[n] = h \sum_{m=-\infty}^{\infty} s[m]g((n-m)T - \tau_d) + v[n]$$

- Suppose that $\tau_d \in [0, T)$ and we want to recover $s[n]$ from $y[n]$

$$y[n] = \underbrace{hs[n]g(\tau_d)}_{\text{desired}} + \underbrace{\sqrt{E_x}h \sum_{m \neq n} s[m]g((n-m)T - \tau_d)}_{\text{ISI}} + \underbrace{v[n]}_{\text{noise}}$$

  - Sampling at the incorrect point creates intersymbol interference!
  - Need *symbol synchronization or equalization* to compensate

Symbol synchronization



ISI created by sampling at the incorrect point

ISI can be eliminated using equalization or sequence detection

○ Sampling with ISI

○ Sampling without ISI

Example with BPSK using a sinc pulse shaping filter

Motivation for frame synchronization

- Suppose that $\tau_d = dT$ for some integer d

$$
\begin{aligned}
y[n] &= h \sum_{m=-\infty}^{\infty} s[m] g((n-m)T - \tau_d) + v[n] \\
&= h \sum_{m=-\infty}^{\infty} s[m] g((n-m)T - dT) + v[n] \\
&= hs[n-d] + v[n].
\end{aligned}
$$

  - Sampling at integer offsets creates unknown delay
  - Use *frame synchronization* to find the beginning of thedata

Motivation for channel estimation and equalization

- Suppose that synchronization has been performed, effectively $\tau_d = 0$

$$y[n] = hs[n] + v[n]$$

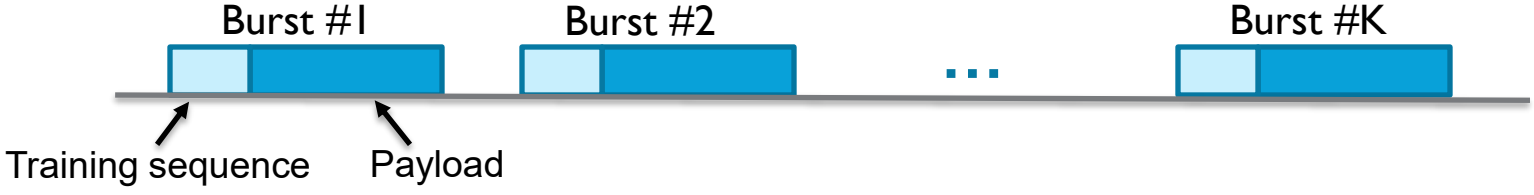
  - Sampling and removing delay leaves distortion due to α and ϕ
  - Some modulation schemes are robust to such distortion (e.g., differential QPSK)
  - For general linear modulation, need *channel estimation and equalization*

Mitigating channel distortions

- General algorithmic approaches

Approach used in this class

  - *Burst mode* - operating on bursts or frames of data, often applied at start of data collection, estimate and correct for impairment

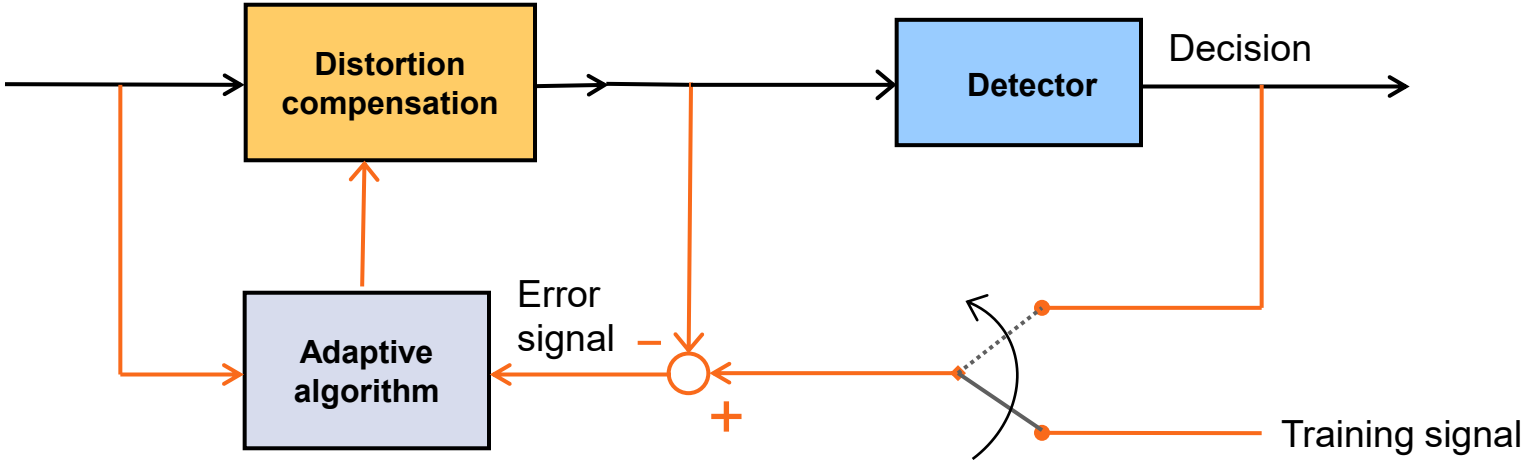


  - *Continuous mode* - operating on continuous data acquisition, often applied to adaptively compensate for distortions



Mitigating channel distortions

- A combination of both approaches is possible
  - Frame headers or training sequences may be used to make an initial correction
  - A decision directed data mode may be used to adaptively compensate for changing distortions



Summary

$$y[n] = h \sum_{m=-\infty}^{\infty} s[m]g((n-m)T - \tau_{\mathrm{d}}) + v[n]$$

- Flat channel model
  - τ_d → Symbol timing mismatch; Frame timing mismatchs → **Symbol synchronization Frame synchronization**
  - h → Amplitude distortion → **Channel estimation and equalization**

Challenge of communication in frequency flat channels

Learning objectives:

- Explain different impairments caused by the frequency flat wireless communication channel

Symbol synchronization (also known as timing recovery)

Learning objectives:

- Formulate the discrete-time symbol synchronization problem
- Solve the symbol synchronization problem using maximum output energy maximization

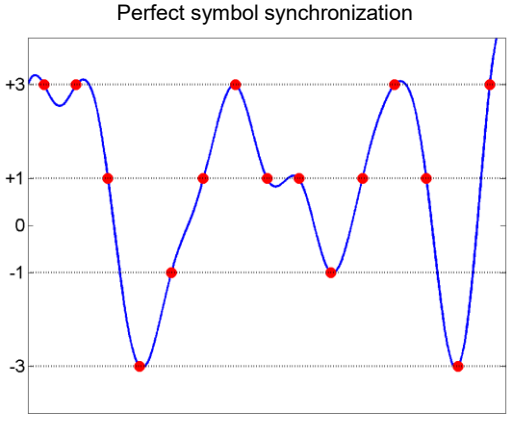
Symbol synchronization

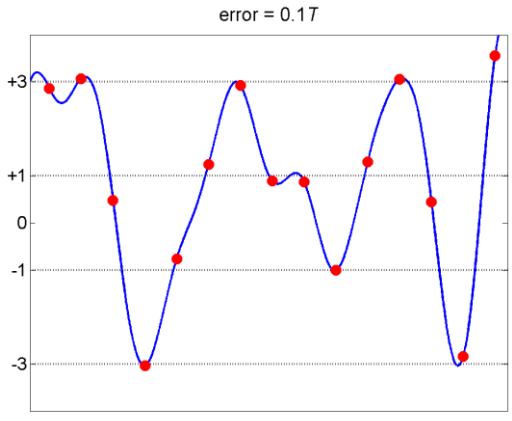
- Frequency flat channel model
- After matched filtering and sampling

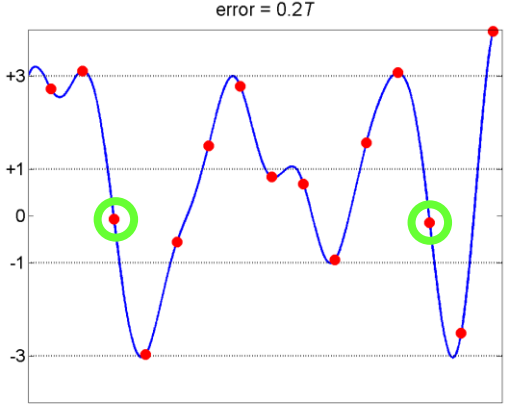
$$y[n] = \underbrace{hs[n]g(\tau_d)}_{\text{desired}} + \underbrace{\sqrt{E_x}h \sum_{m \neq n} s[m]g((n-m)T - \tau_d)}_{\text{ISI}} + \underbrace{v[n]}_{\text{noise}}$$

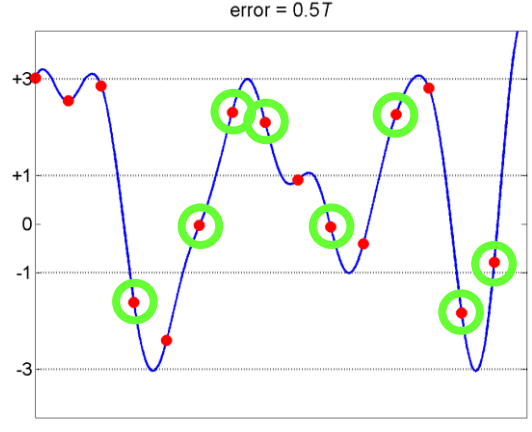
- The purpose of symbol synchronization, or timing recovery, is to **estimate and remove the fractional part of** τ_d within $[0, T)$

Effects of timing error: 4PAM, rolloff 0.15

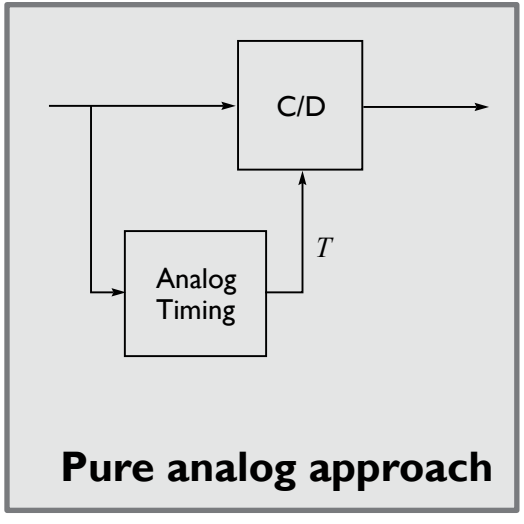




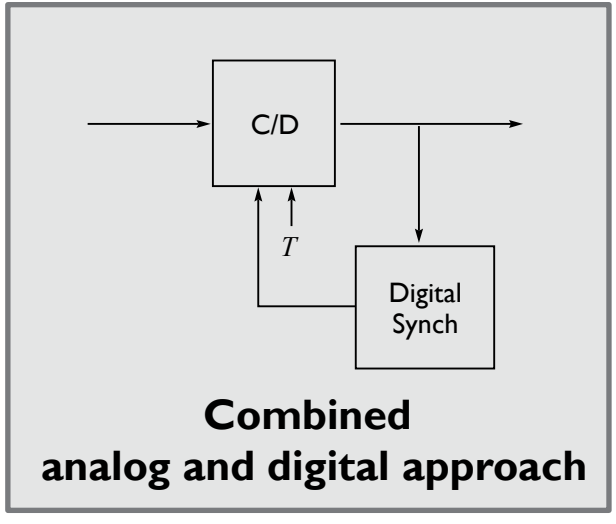




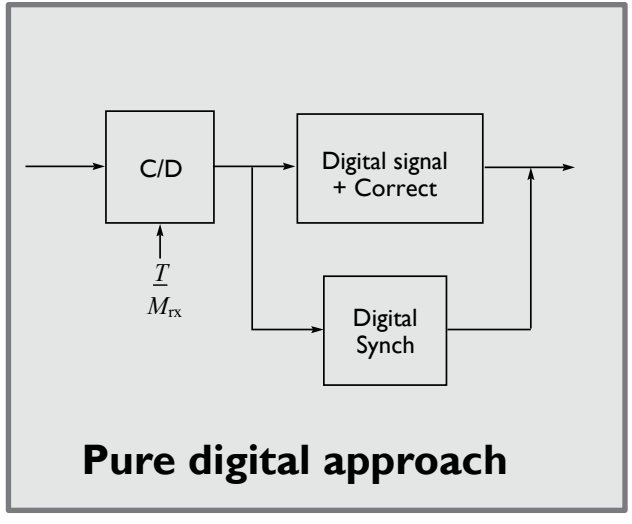
Approaches to symbol synchronization



Pure analog approach



Combined analog and digital approach

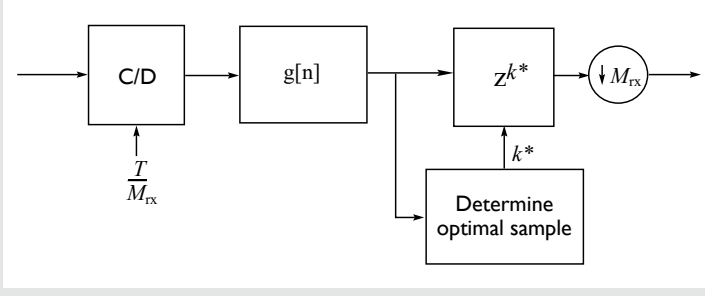


Pure digital approach

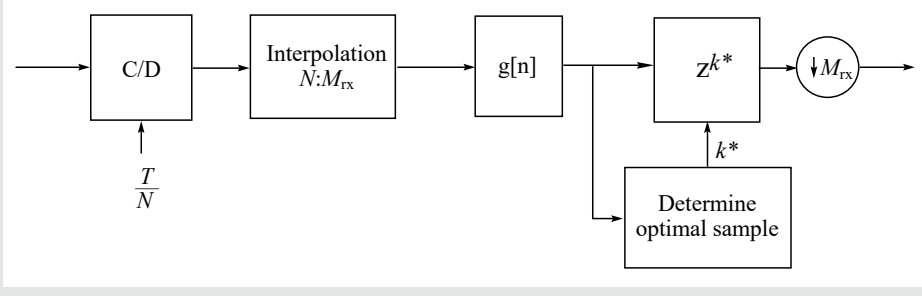
Approach used in this class

Pure digital symbol synchronization

- Assume that $g(t)$ is a Nyquist pulse shape
- Two general digital approaches depending on the sampling rate



Oversampling method

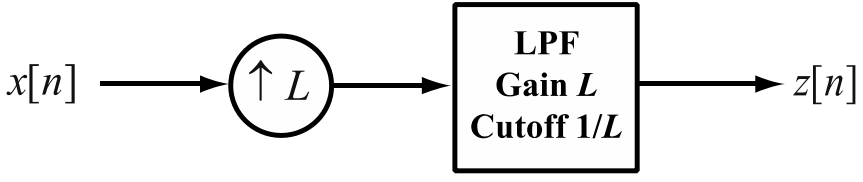


Resampling method

- Oversampling method suitable when M is large
- Resampling or interpolation method suitable when M is small, e.g. $M = 2$

Review of resampling (1)

- Interpolation: increasing the sampling rate by an integer factor L

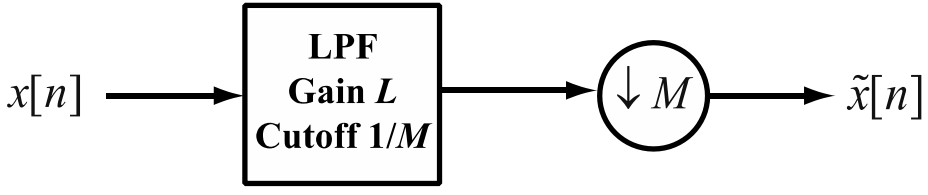


$$x(t) = \sum_m x[m]\,\mathrm{sinc}((t - mT)/T) \longrightarrow x(nT/L) = \sum_m x[m]\,\mathrm{sinc}((nT/L - mT)/T)$$

$$= \sum_m x[m]\,\mathrm{sinc}((n - mL)/L).$$

Review of resampling (2)

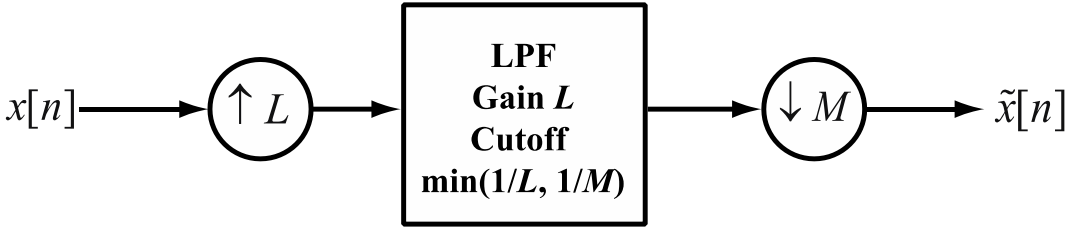
- Decimation: decreasing the sampling rate by an integer factor M



$$\tilde{x}[n] = M \sum_{m} x[m] \operatorname{sinc}((nM - m)/M)$$

Review of resampling (3)

- Changing the Sampling Rate by a Rational Factor



$$R = \max(L, M) \quad \longrightarrow \quad \tilde{x}[n] = \frac{L}{R} \sum_{m} x[m] \, \mathrm{sinc}((nM - m)/R)$$

Maximum output energy criterion (1)

- The best approach would be to formulate and solve for the maximum likelihood symbol timing estimate
  - Somewhat complicated in practice as it requires a search
- Output maximization is one criteria for symbol timing that has a simpler implementation
  - Consider the continuous-time output of the matched filter $y(t)$ followed by sampling at $nT + \tau$

$$J_{\mathrm{MOE}}(\square) = \mathrm{E}\left[|y(nT + \square)|^2\right]$$

Maximum output energy criterion (2)

- The maximum output energy (MOE) solution is

$$\hat{\tau}_d = \arg\max_{\tau \in [0,T)} J_{\mathrm{MOE}}(\tau)$$

- Why does this work? (for most pulse shapes…)

$$\mathbb{E}\left[|y(nT+\tau)|^2\right] = |h|^2 \sum_m |g(mT+\tau-\tau_{\mathrm{d}})|^2 + N_{\mathrm{o}}$$
$$\leqslant |h|^2 |g(0)|^2 + N_{\mathrm{o}}$$

The unique maximum of $J(\tau)$ corresponds to $\tau = \tau_d$

Maximum output energy criterion (3)

- Non adaptive MOE estimator
  - Let $r[n]$ be the output of the matched receiver filter prior to downsampling

$$\tilde{y}[n] = \sum_m r[m] g_{rx}[n-m]$$

  - Approximate the cost function replacing the expectation with a time average over P symbols

$$J_{MOE,d}[k] = E\left[|\tilde{y}[nM+k]|^2\right] \longrightarrow J_{MOE,e}[k] = \frac{1}{P}\sum_{p=0}^{P-1} |r[pM+k]|^2$$

  - The solution is $k^* T/M_{rx}$, with k^* maximizing $J_{\mathrm{MOE,e}}[k]$

Maximum output energy criterion (4)

- Adaptive MOE estimator

1. Differentiating the expectation and approximating

$$\frac{d}{d\square} J_{MOE}(\square) \simeq E\left[\frac{d}{d\square}|y(nT+\square)|^2\right]$$

$$= E\left[y(nT+\square)\frac{\delta}{\delta\square}y^*(nT+\square)\right] + E\left[y^*(nT+\square)\frac{\delta}{\delta\square}y(nT+\square)\right]$$

$$= 2\mathrm{Re}\left[E\left[y(nT+\square)\frac{\delta}{\delta\square}y^*(nT+\square)\right]\right].$$

Maximum output energy criterion (5)

- Adaptive MOE estimator (contd.)

2. Discrete-time difference

$$\frac{\delta}{\delta\tau}y^*(nT+\tau) \simeq y^*\left(nT_s+\tau+\delta\right) - y^*\left(nT_s+\tau-\delta\right)$$

3. Replace $E\{\cdot\}$ by time-average

$$\frac{\mathrm{d}}{\mathrm{d}\square}J(\square) \simeq \frac{1}{P}\sum_{p=0}^{P-1} 2\mathrm{Re}\left[y\left(pT+\square\right)\left(y^*\left(pT+\square+\text{″}\right) - y^*\left(pT+\square-\text{″}\right)\right)\right]$$

Maximum output energy criterion (6)

- Adaptive MOE estimator (contd.)

4. - Assume oversampling or interpolation, being $r[n]$ the output of the matched receiver filter with M samples per symbol period
   - Choose δ a multiple of T/M

$$J_{\mathrm{MOE,der}}[k] = \sum_{p=0}^{P-1} 2\mathrm{Re}\left[\tilde{y}[pP+k]\left(\tilde{y}^*[pP+k+1] - \tilde{y}^*[pP+k-1]\right)\right]$$

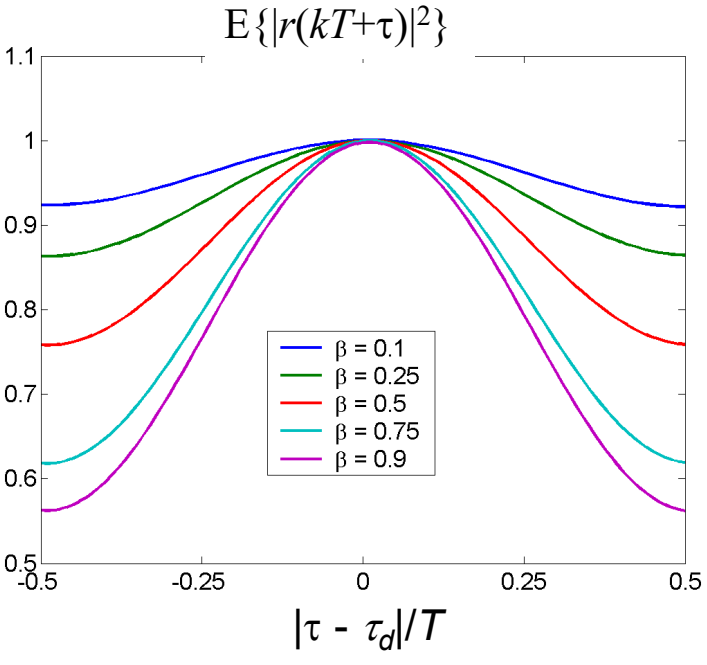
- For $k=0,\ldots,M_{\mathrm{rx}}-1$, the solution is k^* that makes $J_{\mathrm{MOE,der}}[k]$ as close as possible to zero

$$\hat{\tau}_d = k^* T/M$$

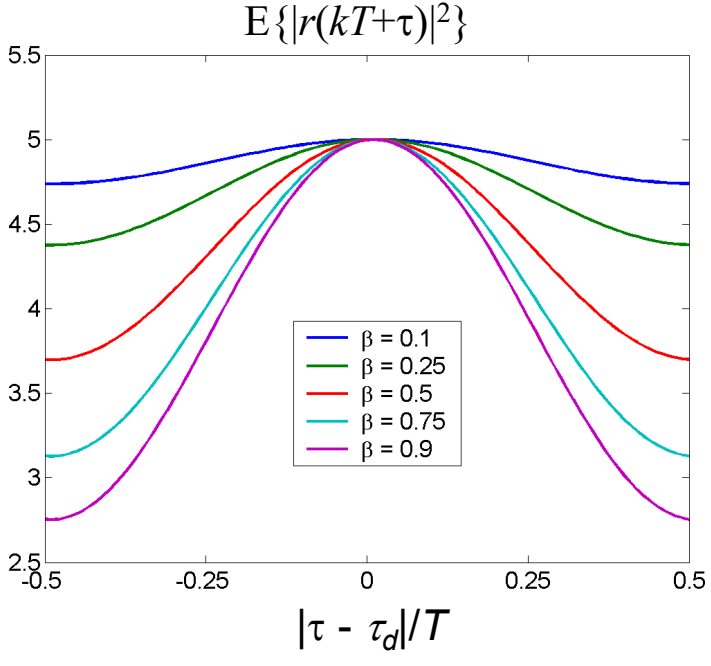
Maximum output energy criterion (7)

- $\mathrm{E}\{|r(kT+\tau)|^2\}$ is maximum when the timing error is zero

BPSK modulation

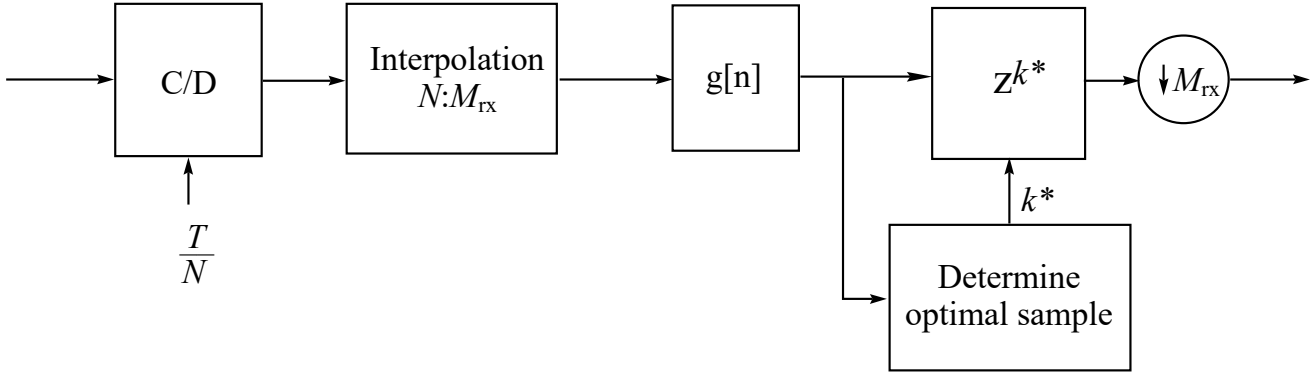


4-PAM modulation



Maximum output energy criterion (8)

- Once an estimate of the offset $\hat{\tau}_d$ is obtained, then it is possible to incorporate the delay into the discrete-time matched filter
  - Processing complexity is reduced



Impact of estimation error on BER performance

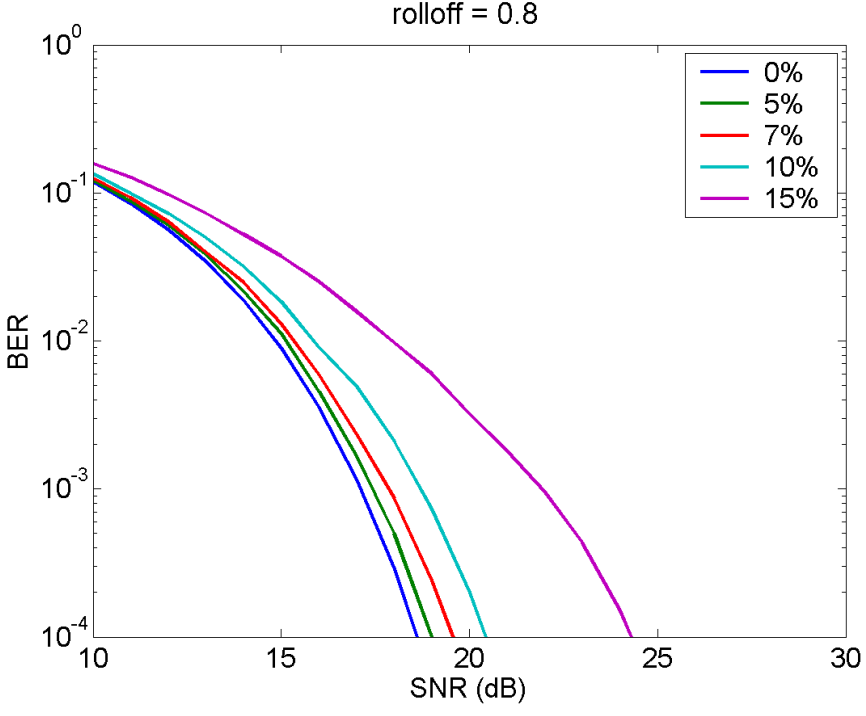
- For pulse amplitude modulation and using the union bound

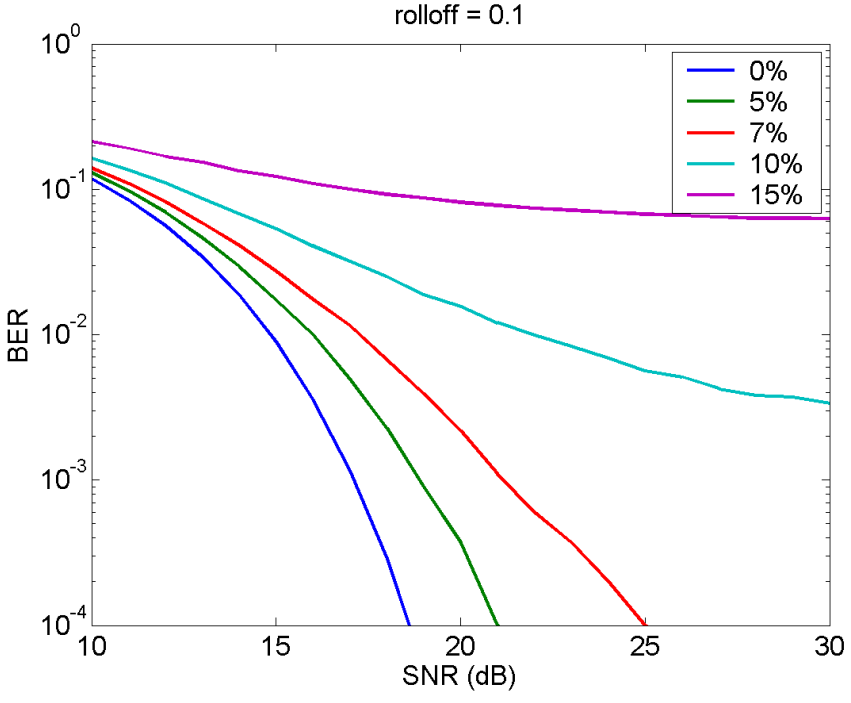
$$\mathrm{SINR} = \frac{|h|^2 \left| g\left(\frac{T}{2M_{rx}}\right)\right|^2}{|h|^2 \sum_{m\neq 0} \left| g\left(mT + \frac{T}{2M_{rx}}\right)\right|^2 + N_0}$$

$$P_e \leq (M-1)\, Q\left(\sqrt{\frac{1}{3M-1} \cdot \frac{|h|^2 \left| g\left(\frac{T}{2M_{rx}}\right)\right|^2}{|h|^2 \sum_{m\neq 0} \left| g\left(mT + \frac{T}{2M_{rx}}\right)\right|^2 + N_0}}\right)$$

Impact of estimation error on BER performance

- 4-PAM, bit error rate for several timing errors $|\tau - \tau_d|/T$





Symbol synchronization or timing recovery

Learning objectives:

- Formulate the discrete-time symbol synchronization problem
- Solve the symbol synchronization problem using maximum output energy maximization

Summary

- Flat channel model $h_c(t) = \square\delta(t - \square_d)$

  - τ_d → Symbol timing mismatch; Frame timing mismatch → **Symbol synchronization, Frame synchronization**

  - α → Amplitude distortion → **Channel estimation and equalization**

Next lecture